JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 101, NO. B1, PAGES 601–612, JANUARY 10, 1996

Magnetostratigraphic dating of the uplifted atoll of Maré: Geodynamics of the Loyalty Ridge, SW Pacific

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Abstract. The Loyalty Islands (SW Pacific) are uplifted as they are progressively affected by the lithospheric flexure of the Australian plate, before its subduction under the New Hebrides Arc. These geodynamic changes are constrained by magnetostratigraphically dating two sections from Maré Island, where mineral extractions coupled with rock-magnetic experiments suggest that the magnetic remanence is mostly carried by a mixture of single-domain to multidomain magnetite/maghemite. With the help of faunal determinations and radiometric dating, the sequences of polarity reversals, correlated to the geomagnetic polarity timescale, range from the top of Chron C4n (late Miocene) to the top of the Gauss Chron (late Pliocene). This new chronostratigraphy refines the timing of two distinctive carbonate units (rhodolith platform/coral reefs) separated by a hardground whose transition is known to coincide approximately with a regional event around the Miocene/Pliocene boundary. The magnetostratigraphic dating indicates that the hardground represents about a 1.9 m.y. hiatus and suggests variable sedimentation rates ranging from 4.7 to 65.4 m/m.y. during the atoll construction. The lithospheric bulge seems to have influenced the evolution of Maré Island some 3.1 m.y. ago, leading to a diachronous emersion of the northeast and southwest rim of the atoll with a mean uplift rate of the order of 4 cm/kyr.

Introduction

The subduction of the Australian plate under the Pacific plate is associated with a bending of the Australian oceanic lithosphere west of the New Hebrides trench [*Dubois et al.*, 1974; *Collot et al.*, 1975]. The modeling of this lithospheric bulge is mostly based on data from uplifted atolls (Loyalty Islands) and seamounts (Bougainville, Sabine Bank) that cover the plate west of the trench [*Dubois et al.*, 1974, 1977, 1988; *Marshall and Launay*, 1977; *Gaven and Bourrouilh-Le Jan*, 1981; *Carrière*, 1987a, b; *Montaggioni et al.*, 1991; *Collot et al.*, 1992]. Related to their motion over the bulge, the geodynamic instability of the Loyalty Islands is reflected by the successive tilting of the carbonate units, recorded within their sedimentary and diagenetic evolution [*Chevalier*, 1968; *Carrière*, 1987b; *Aïssaoui*, 1988]. Those islands consist of volcanic basement overlain by two different sequences of partially dolomitized carbonates, ranging from the Miocene to the present [*Chevalier*, 1968, 1973]. The lower sequence is a platform-type accumulation of rhodoliths; the second is made of reef materials. Similar and exclusive rhodolith sedimentation is reported from various locations in the SW Pacific, in the Far East and in the Caribbean [*Bourrouilh-Le Jan*, 1979; *Garrabé*, 1983].

Regarding their potential to record eustatic, climatic, and tectonic variations, the Loyalty Islands are of prime interest for sedimentologic and geodynamic investigations. However, the study of their evolution remains often qualitative, since their time frame is not precisely known due to the rarity of biostratigraphic markers and to the limitations of radiometric dating of shallow-water carbonates. Paleomagnetic studies on the Bahamas platform [*McNeill et al.*, 1988; *McNeill*, 1990], on Mururoa atoll [*Aïssaoui et al.*, 1990; *Aïssaoui and Kirschvink*, 1990], and on an Australian platform [*McNeill et al.*, 1993] demonstrated that stable remanent magnetization can be preserved in Cenozoic shallow-water carbonates despite their weak magnetization and their diagenetic alteration, including dolomitization.

In this paper, we report magnetostratigraphic dating of two sections from Maré Island (21.5°S, 168.0°E). To test the potential of magnetostratigraphy on late Cenozoic carbonates, related both to eustatic and geodynamic influences, Maré was selected, since its sedimentologic and diagenetic features have been well established previously [*Chevalier*, 1968; *Carrière*, 1987a, b; *Aïssaoui*, 1988]. Furthermore, Maré is the most uplifted atoll of the Loyalty Islands, allowing the best recovery of the sedimentary pile, especially the rhodoliths, which barely outcrop on other islands. Our objectives are to document the timing of the atoll construction and to discuss its evolution related to an active plate boundary.

Regional Context and Geology

The Loyalty Ridge is a 1000-km-long submarine volcanic chain, separated by the Loyalty basin from the island of New Caledonia some 100 km west. Oriented parallel to each other along the 310°N direction, these structures belong to the eastern margin of the Indo-Australian plate. They are bounded to the east by the New Hebrides deep trench oriented 340°N, the site of the subduction (Figure 1a). At 20°S the Australian and Pacific plates have a relative convergent motion of 10 to 12

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Paper number 95JB02878
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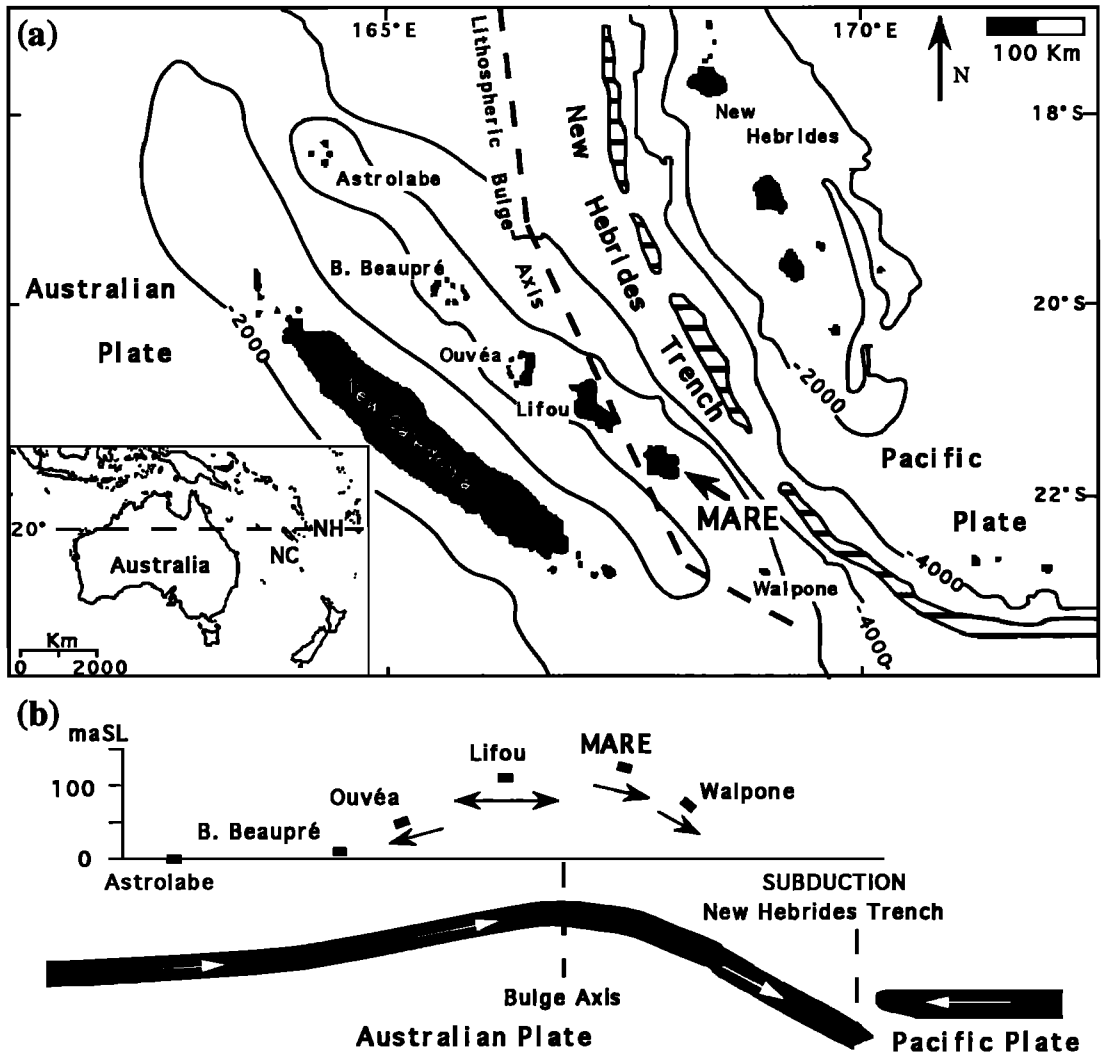


Figure 1. (a) Location of Maré atoll and regional geodynamics; NC, New Caledonia; NH, New Hebrides. (b) Simplified profile of the deflection of the oceanic lithosphere prior to subduction, showing the elevation of the Loyalty uplifted atolls in respect to their position on the lithospheric bulge. Black arrows correspond to the present-day tilting of the islands.

cm/yr along the 75°N direction [*Minster and Jordan*, 1978; *Louat and Pelletier*, 1989]. The Loyalty Islands are uplifted atolls over intraplate basaltic volcanoes, the basalts of which are observed only on Maré Island. From the northwest to the southeast, the elevation of the atolls increases in correlation with their proximity to the lithospheric bulge (Figure 1b). The coral reefs being previously near sea level, the correlation between the altitude of the atolls and their proximity to the trench indicates the vertical amplitude of the bulge, its summit being located between Lifou and Maré, close to Maré [*Dubois et al.*, 1974] and parallel to the trench [*Collot et al.*, 1975].

Maré is presently surrounded by living fringing reefs and has an atoll morphology with a peripheral rim and a central depression (Figure 2). The island consists of a volcanic basement overlain by two lithologically distinct carbonate sequences partially dolomitized, which are uplifted to a maximum of 138 m above sea level and tilted to the NE. Following the end of the volcanic activity, the basaltic substratum was subjected to marine erosion, and fringing reefs colonized its flanks, as shown by the presence of remobilized basalt blocks within the carbonate sediments [*Koch*, 1958]. In front of the reefs, an exclusive rhodolith sedimentation developed and progressively buried the volcano almost to the top, forming a vast platform. Rhodoliths are sub-spherical nodules, 5 to 10 cm in diameter, composed of red algae (Rhodophyte) and encrusting foraminifera laminations [*Bosellini and Ginsburg*, 1971]. Their formation implies shallow, high-energy hydrodynamic conditions, their wide lateral extension, so as to create a platform morphology with stratifications dipping seaward on the edge of the platform, suggests an accumulation on a level surface. The accumulation of rhodoliths, mixed within an *Amphistegina*-rich bioclastic sand, is uniformly dolomitized and outcrops for a thickness of 65 m [*Carrière*, 1987a; *Bourrouilh-Le Jan*, 1979]. Based on the analogy with the slope of Mururoa volcano, the rhodolith unit at the platform periphery may be 300 to 350 m thick [*Carrière*, 1987a].

The top of the rhodolith unit is marked by a sharp, "knife-cut," perforated hardground (highly cemented surface representing a hiatus in deposition), which is restricted to a peripheral ring around the island; its sharpness and extension are incompatible with a karstic erosion. Calculations of the paleoobliquity of the marine terraces [*Carrière*, 1987a], together

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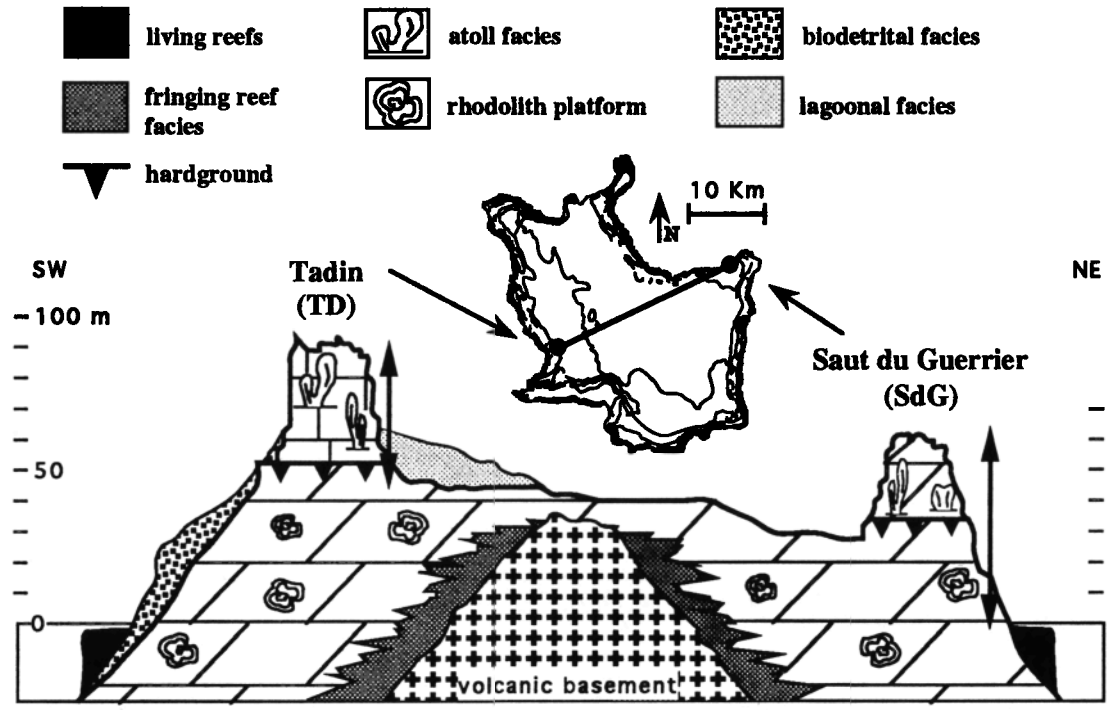


Figure 2. Schematic morphology of Maré Island showing the position of the two studied sections and simplified geologic profile (arrows represent sections).

with geopetal sediments infilling the hardground cavities and perforations, indicate that the hardground was a nearly paleohorizontal datum during its formation. In the central part of the island, no hardground occurs. However, on the inner side, at the base of the reefal rim, large overturned hardground blocks suggest the erosion of the central part of the platform, resulting in a central depression (about 10 m deep) and a peripheral rim [*Aïssaoui*, 1988].

When the sedimentation resumed, a true atoll developed with a reefal crown above the pre-existing rhodolith rim. Lagoonal sediments partially infilled the central depression, the rhodoliths outcropping currently in the central and eastern part of Maré. The atoll rim, dolomitic only in the NE half of the atoll, consists mostly of branched or lamellar corals, either as debris or in growth position, and of biodetritic sediments. The reefal rim is generally thicker in the SW (40 m) than in the NE (20 m). The difference is due to the tilting of the island to the SW as it began to reach the lithospheric bulge [*Carrière*, 1987a, b]. The asymmetry was also pronounced on the outer flanks of the atoll during the uplift phase where small fringing reefs developed in the NE, whereas the sedimentation was mainly biodetritic in the SW. When the atoll passed the summit of the bulge, the tilt direction shifted to the NE [*Carrière*, 1987a, b].

Existing Stratigraphy

At Maré, the age constraints are limited to the volcanic basement, to a few biostratigraphic determinations in the limestones and radiometric dating of reef terraces (Figure 3).

The rhodolith sedimentation started during the late Miocene after the end of the volcanic activity; the terminal emission, based on the outcropping basalt, was dated at 9.3 ± 0.5 Ma by K/Ar [*Baubron et al.*, 1976]. The reefal atoll rim was proposed to be Pliocene (possibly middle Pliocene) at its base, and early Pleistocene at its top [*Chevalier*, 1968]. These ages should be taken cautiously, as they are based on poorly preserved corals [*Chevalier*, 1968] with uncertain biostratigraphic ranges. U/Th dating of aragonitic corals from a +2 to +5 m terrace gave an age of 98,000 ± 15,000 years BP [*Bernat et al.*, 1976] and marine notches at +9 m and +12 m were dated as 125,000 and 205,000 years BP, respectively [*Gaven and Bourrouilh-Le Jan*, 1981].

During this study, several planktic foraminifera were identified (with the help of F.-J. Sierro personal communication, 1993). Only the genus *Globigerinoides*, which first appeared in the basal Miocene and is still living today, is recognized with certainty. Two specimens from about 10 m above the hardground could be either *Globorotalia puncticulata* or *G. inflata* that are distributed from the early Pliocene to present. Specimens located about 40 m below the hardground were determined as either *Globigerinoides extremus* or *G. conglobatus*, whose first appearance occurs during middle Tortonian. These age determinations are in fair agreement with those presented by *Chevalier* [1968] and with the proposed rhodoliths/corals transition age; the disappearance of the rhodoliths and its systematic replacement by corals in the SW Pacific are interpreted as the result of a climatic crisis during the late Miocene-Pliocene [*Bourrouilh-Le Jan*, 1979].

Sections and Sampling

The study is based on two sections, the "Saut du Guerrier" (SdG) section located on the NE side and on the "Tadin" (TD) section at the opposite, SW margin of Maré (Figure 2). This NE-SW direction is parallel to the tilt direction and to the main asymmetric features. For SdG and TD sections, the altitude of the hardground is at +34 and +53 m, respectively. SdG section was sampled from sea level to the top of the atoll rim (+59 m), whereas for Tadin section, the sampling went from 2.5 m be-

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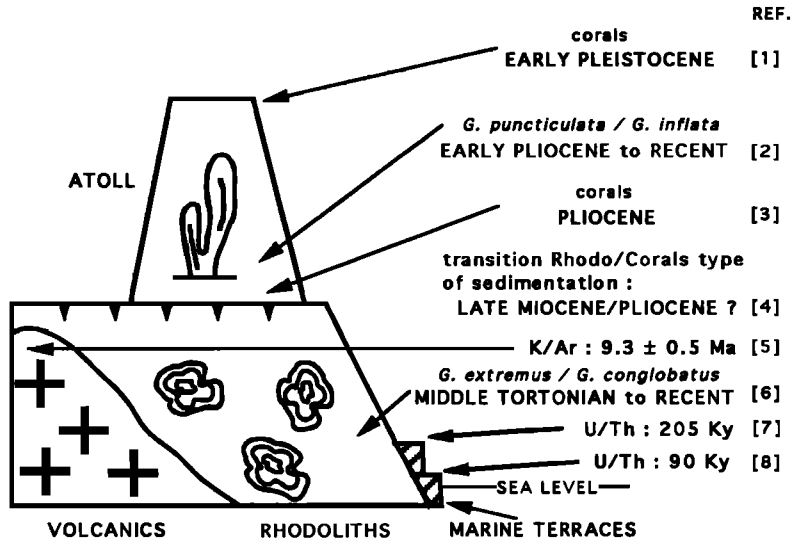


Figure 3. Summary of the radiometric dating and faunal determinations from Maré Island: [1, 3] *Chevalier* [1968]; [2, 6] this study; [4] *Bourrouilh-Le Jan* [1979]; [5] *Baubron et al.* [1976]; [7] *Gaven and Bourrouilh-Le Jan* [1981]; [8] *Bernat et al.* [1976].

low the hardground to the top of the reefal unit (+93 m). In both sections the mineralogy of the rhodolith sediments is dolomite. The atoll rim mineralogy is dolomite at SdG and low-magnesium calcite at TD. For SdG and TD sections, no major hiatuses other than the hardground separating the rhodoliths from the reefs are reported in the literature, nor were any observed during the field sampling.

Oriented paleomagnetic plugs, 3-4 cm long and 2.5 cm in diameter, were drilled from the outcrop with a nonmagnetic diamond core bit. Prior to measuring, they were immersed for 30 s in a 0.1 *N* HCl solution and cleaned in distilled water to remove possible contamination during the drilling phase. Two hundred and fifty-nine samples were collected from the SdG section, averaging one sample per 23 cm. In the TD section, 176 samples were taken at an approximately 24-cm interval, including a nonrecovered interval of 5.8 m between +58.1 and +63.9 m due to nonexposure.

Measurements

Procedure

The samples were measured with a 2G superconducting magnetometer in the magnetically shielded room at the Rosenstiel School of Marine and Atmospheric Science, University of Miami. Specimens were stored in the shielded room for 48 hours prior to the measurement of the natural remanent magnetization (NRM), then demagnetized by alternating field (from 2.5 to 10 mT for TD and up to 15 mT for SdG) followed by stepwise thermal demagnetization in a nonmagnetic furnace (from 100° to 600°C). Initial magnetic susceptibility was measured on all samples using a Bartington MS2 sensor, then monitored on 29 selected specimens after each thermal heating step to check for possible mineralogical transformation. Orthogonal projections of the demagnetization data were used to determine polarity [*Zijderveld*, 1967]. Direction angles were calculated using a leastsquares method [*Kirschvink*, 1980].

Rock-magnetic experiments and magnetic grain separations were carried out on selected samples to identify the carriers of the NRM. Magnetic minerals were extracted following the method described by *Chang and Kirschvink* [1985] and were examined under a transmission electron microscope (TEM JEOL 100CXII) for the ultrafine fraction and scanning electron microscope (SEM Philips 505) for the coarser fraction. Both TEM and SEM examinations were coupled with energie dispersive spectrum (EDS) microprobe for geochemical analyses. Hysteresis loops were acquired using an alternating gradient force magnetometer (Micromag), at the Centre des Faibles Radioactivités of Gif-Sur-Yvette.

Magnetic Intensity

NRM intensities of Maré carbonates are similar to other late Cenozoic shallow-water carbonates [*McNeill et al.*, 1988, 1993; *Aïssaoui et al.*, 1990; *Cunningham et al.*, 1994]. In SdG, they range from 4.3×10^{-10} to 5.9×10^{-6} A m^2/kg (mean 1.4×10^{-7} A m^2/kg), whereas they are weaker in TD, varying from 5.6×10^{-10} to 1.1×10^{-6} A m^2/kg (mean 2.3×10^{-8} A m^2/kg). When compared to the rhodolith specimens they overlie, reefal samples have a 10-20% mean increase in NRM intensity. Limestone specimens (TD reefal unit) have a mean intensity slightly higher than the underlying dolomitized rhodoliths (2.3×10^{-8} versus 2.1×10^{-8} A m^2/kg), but the former are significantly weaker than the dolomitized specimens from the SdG reefal unit. Thus the limestone to dolomite transition in this case is not followed by significant change of the NRM, suggesting the preservation of original remanence through dolomitization [*McNeill et al.*, 1988; *Aïssaoui et al.*, 1990; *McNeill and Kirschvink*, 1993]. Variations of NRM intensity correlate roughly with the susceptibility curves, suggesting that ferromagnetic minerals carry the NRM, although susceptibility data are dominated by dolomite and calcite diamagnetism as indicated by their negative values.

Magnetic Polarity

For the TD section, 124 of the 176 specimens (70%) were used to determine the polarity sequence. For the SdG section, we have used 133 of the 259 measured specimens (51%). The results from 52 (TD) and 126 (SdG) specimens were not inclu-

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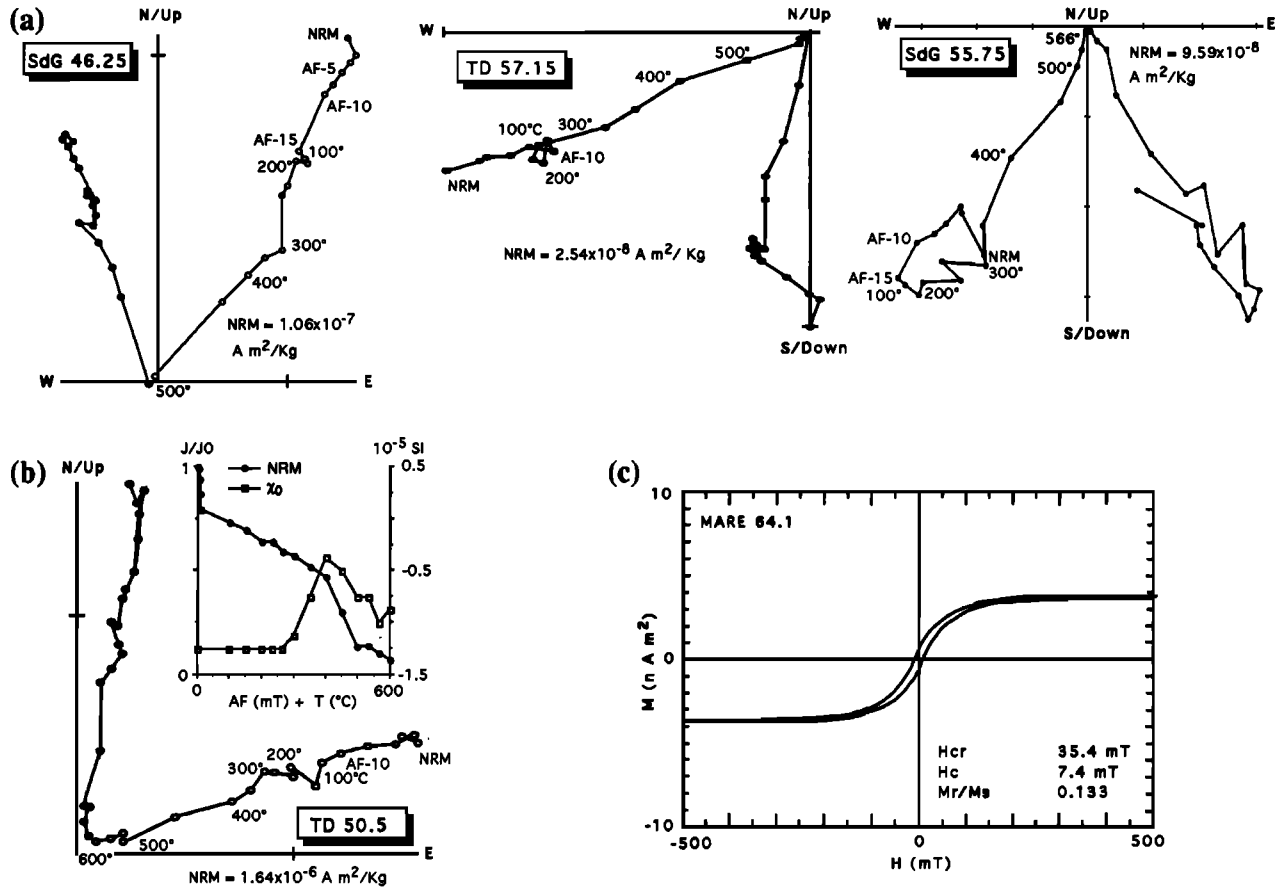


Figure 4. (a) Typical Zijderveld diagrams illustrating normal polarity (SdG46.25) and reversed polarity (SdG57.15 and TD55.75) from Maré samples. (b) Synthesis of the results from sample TD50.5: Zijderveld diagrams; evolution of the NRM during alternating field (AF) (first five points) and thermal demagnetization; evolution of susceptibility (χ_0) during heating. (a and b) Open and solid circles are projections on the vertical and horizontal plane, respectively; the AF demagnetization steps are in mT and the thermal in degrees Celcius. (c) Representative hysteresis loop from Maré carbonates. The raw data have been slope-corrected for the paramagnetic or diamagnetic contribution.

ded because they were either too weak (<2x10^{-9} A m^2/kg), unstable, or did not completely decay towards the origin. Specimens with a mean angular deviation above 15° were also discarded except when they had an unambiguous polarity and when they were consistent with reliable adjacent samples. They were, however, not included in the statistical analysis. Single-point reversals were not used except in one case: at the top of TD where a single-point reversal, which needs additional verification, represents the uppermost sample of the section.

Most of the usable specimens exhibit a single component of magnetization after the decay of a weak viscous overprint (SdG46.25, TD57.15, and TD50.5 in Figure 4). For a few specimens, a secondary component is superimposed on the characteristic component (SdG55.75 in Figure 4). These viscous and secondary components were usually removed during the first steps of alternating-field demagnetization and by heating to 150°-200°C, respectively. Usually, after removal of the secondary component, the high-temperature trajectory shows a stable decay to the origin until 450°-500°C (up to 560°-600°C for 40% of the samples), temperatures at which most specimens lost their stable remanence.

Determination of the sample polarity is based upon the characteristic component directions. Directions used in statistical calculations (Table 1) are plotted in Figure 5. The inclination-only method of *Kono* [1980] gives angles, for both combined polarities, slightly shallower than the expected angle for the geocentric axial dipole for Maré (38°). Fisherian means yielded normal and reversed mean directions close to antipodal, the directions within each polarity class being very scattered, however. The reason for such scattering is unclear, although the weak magnetization and the process of NRM acquisition in shallow-water carbonates (compaction, bioturbation, early cementation) may contribute to it; thus no attempt was made to calculate paleopositions for Maré Island.

Source of Remanent Magnetization

TEM and SEM Investigations

The TEM examinations of the magnetic extracts reveal the presence of fine-grained magnetic crystals (0.02 to 0.25 μm) which are commonly grouped into clusters of up to 50 individual crystals randomly oriented (Figure 6a and 6b). Well-formed crystals with sharp edges showing prismatic and cuboidal morphologies coexist with crystals exhibiting smooth boundaries and rounded shape. These grains, which fall in the single-domain (SD) and two-domain particles range of the theoretical stability diagram of magnetite [*Butler and Banerjee*, 1975], are

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Table 1. Summary of Kono and Fisher Statistics

Section	Method	Component	*N*	Declination, deg	Inclination, deg	*K*	α95, deg
SdG	Kono	General	101		33.2	10.5	4.5
TD	Kono	General	107		30.9	12.8	3.9
SdG	Fisher	Normal	70	8.3	-44.7	4.1	9.7
SdG	Fisher	Reversed	31	189.9	33.1	4.2	14.4
SdG	Fisher	General	101	8.8	-41.2	4.1	8
TD	Fisher	Normal	83	29.8	-37.6	5.6	7.3
TD	Fisher	Reversed	24	180	28.9	4.1	16.8
TD	Fisher	General	107	23.1	-36.4	4.8	7.0

N, number of samples used in statistical calculation; *K*, precision parameter; α95, 95% probablility angular radius.

probably magnetite/maghemite, as they contain only iron (Figure 6c). The fine (SD) magnetic grains extracted from Maré carbonates are similar in morphology and composition to magnetofossils discovered in Cenozoic carbonate settings, where they represent the main carrier of remanent magnetization [*Chang et al.*, 1987; *Vali and Kirschvink*, 1989; *McNeill et al.*, 1988; *McNeill*, 1990; *Aïssaoui et al.*, 1990].

In addition to the fine-grained SD magnetite/maghemite, larger grains (3 to 20 μm) occur in the extracts. Their morphology is mostly irregular, angular, or rounded; few cubic or spherical grains also occur. The large grains contain either iron, or iron with detectable amounts of titanium and, in rare cases, chromium. Their shape and composition suggest that these grains are iron oxides of detrital origin (titanomagnetite/titanomaghemite). The volcanic basement of Maré is probably the main source for the detrital multidomain (MD) grains due to the remote location of the island, far from any terrestrial input. The grains having only pure iron composition could be magnetite/maghemite, although goethite cannot be excluded, even though no sharp loss of intensity occurred around 80°-120°C (Tc of goethite) during thermal demagnetization. Finally, while pseudo-single-domain (PSD) magnetic grains were seldom observed within the Maré extracts, one cannot rule out their presence (separation method, TEM resolution, clustering on larger grains). Such grains are likely present in the samples and therefore contribute to the magnetization.

Hysteresis Loop and Thermal Evolution

Sixteen bulk unheated samples were measured for their hysteresis properties. Only five of them gave coherent loops due to the weak magnetization of Maré carbonates. The hysteresis data indicate the predominance of low-coercivity minerals (29.4<Hcr<40.5 mT) within the range of magnetite/maghemite (Figure 4c). The ratios of saturation remanence to saturation magnetization (0.13<Mr/Ms<0.18) and those of remanent coercivity to the bulk coercivity (3.04<Hcr/Hc<4.78) border the PSD/MD boundary, according to the criteria of *Day et al.* [1977]; they fall above the trend line defined for the SD+MD mixture [*Parry*, 1982], suggesting a more complex grain size distribution. Although in limited number, the Maré carbonate hysteresis data seem to fit better the values of other unremagnetized carbonates [*Borradaile et al.*, 1993; *Channel and McCabe*, 1994; *Belkaaloul et al.*, 1995] than those of remagnetized limestones [*Jackson, 1990*].

Most of the samples display unblocking temperatures below 500°C, the remaining magnetization being lost before 600°C. Together with the coercivity values, this response suggests the presence of magnetite (Tc = 575°C) and the absence of hematite (Tc = 675°C). The behavior of susceptibilities values (χ_o) from most of the 29 selected samples remained stable (i.e., low negative values) during stepwise heating.

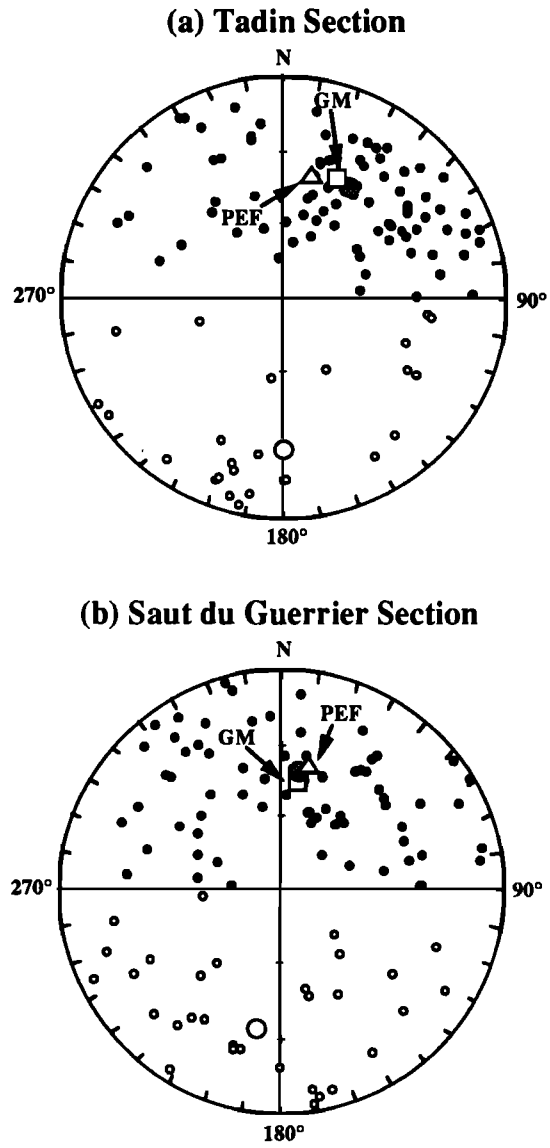


Figure 5. Equal-area projection of characteristic directions used in the statistics for (a) Tadin and (b) Saut du Guerrier sections. Closed (open) symbols, normal (reversed) directions; large solid (open) circle, normal (reversed) mean; GM, general mean; PEF, present dipole field at the sampling site.

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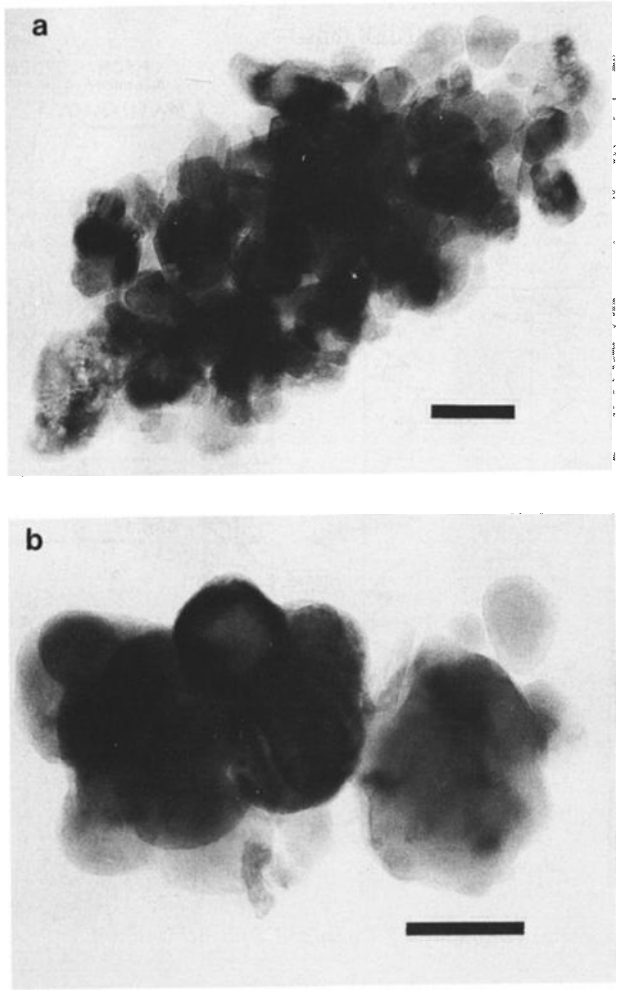


Figure 6. TEM views of SD magnetite minerals extracted from Maré carbonates. Scale is 0.05 μm in both micrographs. (a) Abundant cubic to prismatic magnetite crystals in aggregate; grains are similar in size to SD biogenic magnetite. (b) Small cluster of common single-domain crystals exhibiting rounded shape. (c) Typical energy dispersive spectrum obtained for the grains shown in Figures 6a and 6b.

Five samples showed a slight increase of χ_o in the range 300°-400°C and a decrease before 600°C. These changes were, however, not associated with an intensity jump nor with any directional change of the magnetization trajectory (TD50.5 in Figure 4b). Therefore, in a few samples, a slight mineralogical transformation may occur from a low-susceptibility mineral to a stronger susceptibility one, but it does not mask the characteristic component of magnetization.

In summary, we believe that the characteristic primary magnetization component has been isolated because of the following points

1. The consistency of the magnetic polarity intervals within the two sections.

2. The limited complexity of the magnetic mineralogy of the extracted grains, dominated by a mixture of SD to MD magnetite/maghemite which carry the NRM; the stability over time of SD magnetite/maghemite grains (of probable biogenic origin) likely contribute to the preservation of a legible original magnetization in Maré carbonates.

3. The similarities between the paleomagnetic behavior and the magnetic carriers observed at Maré with those of other Cenozoic shallow-water carbonates.

Polarity and Correlations From SdG and TD Sequences

For the TD and SdG sections, 10 and 13 reversals were found, respectively (Figure 7). The polarity reversals are labeled from Rh1 to Rh6 for the rhodolith unit and A1 to A9 for the atoll unit. The magnetic reversal boundaries (Table 2), placed midway between two adjacent specimens of opposite polarity, are constrained within 0.5 m or less, except for the A3/A4 boundary in the TD section, which occurs in a 5.8-m interval of nonrecovery. Based on a single-point reversal, the A9 interval in the TD sequence needs further confirmation with additional sampling.

With the help of the hardground, which was shown to be near paleohorizontal during its formation, TD and SdG sections are correlated on the basis of reversal pattern similarities (Figure 7). Above the hardground and up to the A6/A7 boundary, the two sections show the same reversal pattern. The A6/A7 boundary is at the same elevation (within 0.35 m) above the hardground. The thickness of each correlated polarity zone does not vary by more than 0.9 m. This implies a somewhat constant rate of sedimentation for these intervals; thus, as a first approximation, we give to A3 in the TD section, the same thickness as its counterpart in the SdG section. In SdG the section ends with the thin reverse zone A7, whereas in TD, A7 is overlain by the thick normal A8, and then by the beginning of the reverse A9. Below the hardground, in the SdG section, the top of the rhodoliths is reverse polarity (Rh6), while it is normal in TD. Since the hardground represents a near paleohorizontal datum, we suggest that the 3 m of normal polarity underlying it in the TD section correspond to the very top of the Rh5 normal interval in the SdG section. Because of its reduced thickness (1.6 m in SdG), the absence of Rh6 in TD may be the result of differential erosion or minor differential topography at the time of formation.

Reversal Chronology

Based on the comparison of our polarity sequences with the geomagnetic polarity timescale [*Cande and Kent*, 1992], the proposed reversal chronology (Figure 7 and Table 2) uses several tie-points (Figure 3), including (1) corals from marine terraces at or below +12 m are not older than 205,000 years [*Bernat et al.*, 1976; *Gaven and Bourrouilh-Le Jan*, 1981]; (2) the base of the reefal rim is not older than Pliocene (possibly middle Pliocene), and its uppermost part is not younger than

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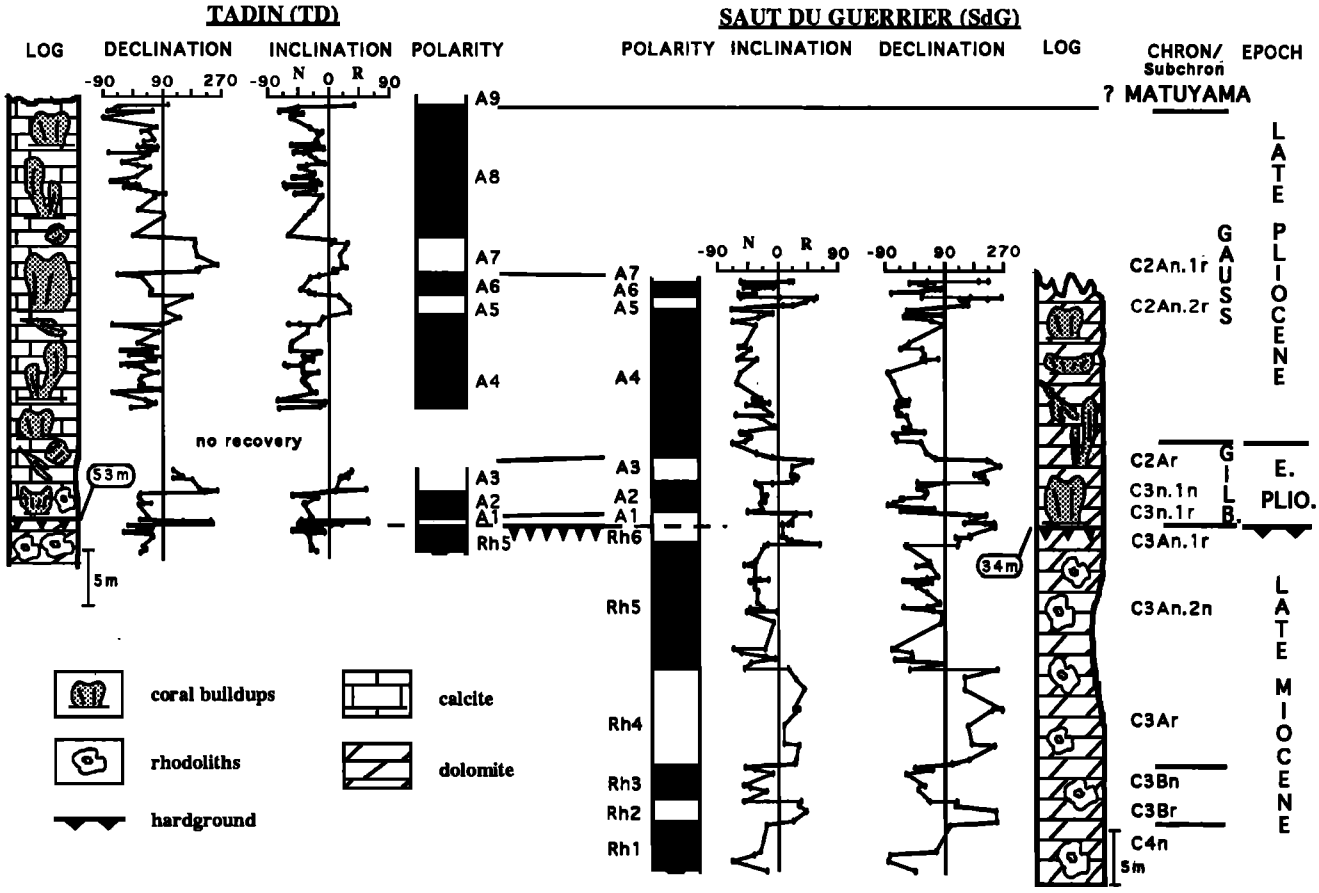


Figure 7. Lithologic log, declination, inclination, and magnetostratigraphy of Tadin and Saut du Guerrier sections.

early Pleistocene [*Chevalier*, 1968]; (3) the rhodolith sedimentation postdates the end of the volcanic activity dated at 9.3 ± 0.5 Ma [*Baubron et al.*, 1976]; and (4) the occurrence of the planktic foraminifera, *Globorotalia puncticula* or *G. inflata* and *Globigerinoides extremus* or *G. conglobatus*, above and below the hardground, respectively, gives a maximum late Miocene age for the upper rhodoliths and an early Pliocene maximum age to the atoll rim. This is in general agreement with the rhodoliths/corals transition related to a regional late Miocene/Pliocene event [*Bourrouilh-Le Jan*, 1979].

Considering the above constraints, we exclude the presence of the Brunhes Chronozone at the top of the atoll rim. We interpret the long normal polarity zone (from A8 to A4) punctuated by two short reversed intervals as the Gauss Chronozone.

Table 2. Summary of the Polarity Interval Boundaries and Dated Horizons Deduced From the Magnetostratigraphy for TD and SdG Sections

Zone	Polarity	Altitude (masl)		Chrons	Age, Ma
		Tadin (TD)	S. Guerrier (SdG)		
A9	R	92.20-		Matuyama (start)	-2.60
A8	N	80.00-92.20		C2An.1n	2.60-3.05
A7	R	76.65-80.00	58.00-	C2An.1r	3.05-3.13
A6	N	74.60-76.65	56.45-58.00	C2An.2n	3.13-3.22
A5	R	72.75-74.60	55.40-56.45	C2An.2r	3.22-3.32
A4	N	58.25-72.75	40.35-55.40	C2An.3n	3.32-3.55
A3	R	55.95-58.25	38.10-40.15	C2Ar	3.55-4.03
A2	N	53.45-55.95	35.15-38.10	C3n.1n	4.03-4.13
A1	R	53.00-53.45	34.00-35.15	C3n.1r	4.13-
Hardground		53	34		
Rh6	R	/	32.40-34.00	C3An.1r	-6.08
Rh5	N	-53.00	20.00-32.40	C3An.2n	6.08-6.38
Rh4	R		10.50-20.00	C3Ar	6.38-6.74
Rh3	N		7.00-10.50	C3Bn	6.74-6.90
Rh2	R		4.75-7.00	C3Br	6.90-7.24
Rh1	N		-4.75	C4n	7.24-

Ages are form *Cande and Kent* [1992].

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Above the Gauss, the reversed polarity zone (A9) may represent the base of the Matuyama Chronozone. Below the Gauss, the three reversals (A3 to A1) are correlated to the upper part of the Gilbert Chron: the reversed C2Ar, the normal C3n.1n and the reversed C3n.1r Subchronozones, respectively.

For the rhodolith unit, two correlations may be possible. The polarity sequence from R6 to R1 could represent (1) Chronozone C3An to the top of Chronozone C4n; (2) Chronozone C4r to Subchronozone C4Ar.2n. The sampled rhodolith section has a thickness of 34 m. This represents a thin, upper part of the total thickness of the rhodoliths accumulation, which could be as thick as 300-350 m [*Carrière*, 1987a]. Since sedimentation begun 9.3 ± 0.5 m.y. ago, the correlation of the basal normal Rh1 with Subchron C4Ar.2n (9.43-9.49 Ma) would leave little time for the remaining hundreds of meters of rhodoliths to accumulate, and would imply rates that are of an order of magnitude faster than for the upper 34 m. Thus we favor the first interpretation but recognize the weakness of this correlation. Therefore, moving downwards from Rh5 to Rh1, we interpret the sequence as the following: normal C3An (incomplete), reversed C3Ar, normal and reversed C3B, and upper part of normal C4n. Just below the hardground, the short reversed Rh6 is interpreted as part of Subchronozone C3An.1r within the normal C3An.

Magnetostratigraphic dating from TD and SdG provides several dated horizons and subsequent accumulation curves (Figure 8). For the rhodolith as well as for the reefal deposits, variable rates of sedimentation ranging between 4.7 and 56.6 m/m.y. are proposed. The upper 30 m of rhodoliths accumulated in about 1.1 m.y. during the very latest Miocene, between 7.2 Ma (base of Subchron C3Br) and 6.1 Ma (base of Subchron C3An.1r). The Miocene/Pliocene boundary is missing, possibly included within the hardground hiatus. Based on the accumulation curves, the hiatus spans between about 6.04 and 4.17 Ma. Following this 1.9 m.y. period of nondeposition/erosion, the reefal sedimentation resumed synchronously in both sections. Unlike its initiation, the atoll rim construction seems to have ended diachronously; at about 3.1 Ma (shortly after C2An.2n/C2An.1r boundary) in SdG and later in TD, at about 2.6 Ma (shortly after Gauss/Matuyama boundary).

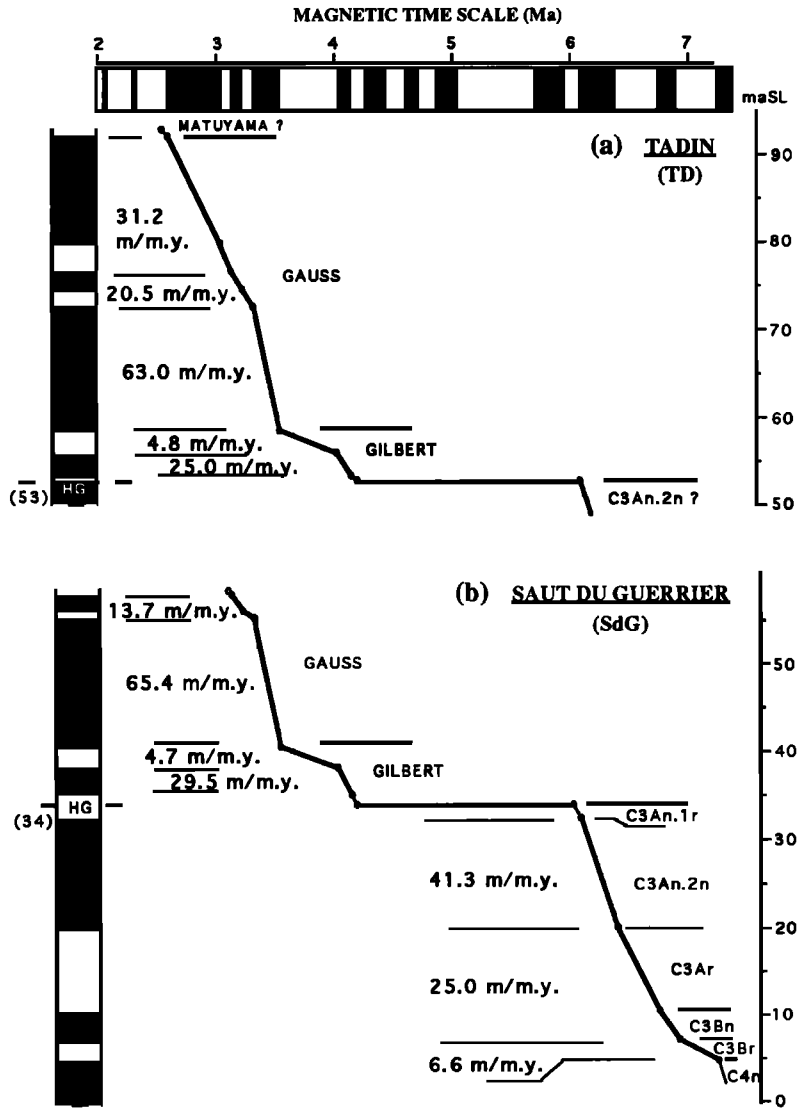


Figure 8. Age/depth curves for (a) Tadin and (b) Saut du Guerrier sections based on magnetostratigraphy [*Cande and Kent*, 1992]. Note that within the two locations, the sedimentation rates were comparable from the beginning of the reefal construction to a mid-Gauss time.

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Thus the 24-m-thick atoll rim from SdG would have formed in about 1.1 m.y., while the 34-m-thick rim from TD was constructed in about 1.6 m.y. The top of the lower Pliocene, at the Gauss/Gilbert boundary, would be at about 6 to 7 m above the hardground, while the Pliocene/Pleistocene boundary was not recorded in the sections.

Discussion

Geodynamics of Maré

Based on the proposed reversal chronology for TD and SdG sections, the evolution of each side of Maré Island seems to have been somewhat similar from the end of the volcanic activity (9.3 ± 0.5 Ma) to about 3.1 Ma (Figure 8). This interval corresponds to the phase during which the island slowly subsided, allowing a uniform accumulation of the rhodoliths, then following the hiatus of 1.9 m.y., the initiation of the reefal buildup. At about 3.1 Ma, the island began its differentiation as it approached the lithospheric flexure. Closer to the bulge, the NE side of the atoll started to be affected by the uplift, leading to a preferential subsidence and tilting toward the SW side. As the coral buildup progressively ceased, the NE side of the atoll then emerged, while sedimentation continued on the SW side. This phase of Maré evolution can be visualized at present by examining Ouvéa atoll, where the east rim is above sea level, while the west rim is still immersed and growing. At about 2.6 Ma, the western side of Maré also began to emerge, progressively leading to the current uplifted atoll.

Based on radiometric dating of aragonitic corals within the terraces, existing uplift rates for the Loyalty Islands indicate that uplift decreases to the west of Maré [*Marshall and Launay*, 1978]. U/Th dating of the 2- to 5-m uplifted terrace from Maré indicate rates about 17.5 ± 1.5 cm/kyr [*Bernat et al.*, 1976]. Based on isotopic stratigraphy of Holocene and Pleistocene marine notches [*Gaven and Bourrouilh-Le Jan*, 1981; *Bourrouilh-Le Jan*, 1985], a much lower rate, ranging between 2.4 and 5.2 cm/kyr, was found. Since the corals were previously near sea level, their age at the top of the atoll rim gives an estimate of the time at which the uplift initiated. This age may be slightly older than the actual uplift age due to the dissolution of reefal facies. However, many of the original rim features (e.g., faros walls) are still well defined today, suggesting relatively low rates of dissolution. Based on corals collected from the top of the emerged crown [*Chevalier*, 1968], the uplift began during the early Pleistocene [*Dubois et al.*, 1974; *Marshall and Launay*, 1978; *Carrière*, 1987a]. Magnetostratigraphic dating of the TD and SdG sections provides new ages for the top of the atoll rim, suggesting that Maré has been influenced by the lithospheric bulge earlier than previously thought. Taking an age of about 3.1 Ma for the corals at the top of SdG the section (+58 m) would lead to an average uplift rate of 1.9 cm/kyr. Tilted towards the NE, Maré has already passed over the top of the bulge, and its NE side (i.e., the SdG section) is presently dipping towards the New Hebrides trench. The uplift rate calculated from the SdG section represents the combination of the uplift and initial subduction, and should thus be discarded. A more realistic rate is obtained through data of the SW rim, i.e., TD, that is closer to the bulge crest. Using an age of about 2.6 Ma for the top of TD (+93 m), the calculated mean rate is 3.6 cm/kyr. This value is a minimum, since the atoll rim culminates on its western and southern sides at altitudes above 100 m. The maximum elevation is at the southern tip of Maré at Wabayoce, which reaches +138 m, with atoll rim 20 m thicker than TD; the corals on top of it are likely to be younger than 2.6 Ma. Finally, the uplift rate of 3.6 cm/kyr represents the average over the last 2.6 m.y. and does not consider variations through time. However, it seems to support the idea of slow uplift.

Regional Correlations

There is no existing shallow-water magnetostratigraphy record for regional correlation in the New Caledonia-New Hebrides area. Paleomagnetic investigation of the Queensland Plateau, Australia, during Ocean Drilling Program leg 133, revealed clear similarities with the present study in regards to the magnetic behavior and magnetic composition of the sediments [*McNeill et al.*, 1993]. Unfortunately, no correlation can be established, since the magnetostratigraphy of the Australian platform went back only to the Matuyama/Gauss boundary. On a broader scale, hiatuses related with sea-level fluctuations are known to have occurred in many locations from the Pacific Ocean [*Lincoln and Schlanger*, 1987; *Ludwig et al.*, 1988; *Aïssaoui and Kirschvink*, 1990; *Montaggioni et al.*, 1991; *Ohde and Elderfield*, 1992; *Aharon et al.*, 1993], at or just below the late Miocene-Pliocene boundary. On Niue atoll, in the South Pacific, prominent dissolution unconformities which started at 6.14 Ma and culminated at 5.26 Ma were dated by Sr-isotope chronostratigraphy [*Aharon et al.*, 1993]. On the atoll of Mururoa, in the Tuamotu Archipelago, the magnetostratigraphy of the top 200 m of sediments shows clear variations of the accumulation rate, with low rates coinciding with hiatal surfaces or subaerial exposures [*Aïssaoui et al.*, 1990; *Aïssaoui and Kirschvink*, 1990]. The Miocene-Pliocene boundary was associated with an unconformity leading to reduced rate of accumulation between 6.0 and 4.6 Ma [*Aïssaoui and Kirschvink*, 1990]. A rate of 57.1 m/m.y. was calculated during lower to middle Gauss, followed with somewhat slower rates for the upper Gauss and the Matuyama. Although far away, the polarity sequences and accumulation rates from Maré and Mururoa show similarities, suggesting variations on a larger scale. However, detailed comparisons are difficult, since Mururoa is a subsiding atoll, while Maré is on an active margin.

On Maré Island, the hiatus that masks the Miocene-Pliocene boundary, is characterized by a sharp sedimentologic transition from rhodoliths to coral reefs. This transition is believed to reflect climatic variations [*Bourrouilh-Le Jan*, 1979]. The proliferation of rhodoliths is related to a colder climate, which did not allow coral colonization during the middle and upper Miocene in the W/SW Pacific. A warming of the climate during late Miocene-Pliocene would then have favored the development of corals. Based on the Maré sequences, we have no arguments either to infer or support a climatic crisis relationship. However, by dating precisely for the first time the rhodoliths/corals transition between 6.04 and 4.17 Ma, the present study confirms a major event, possibly of global scale, at the Miocene-Pliocene transition.

Conclusion

The present study reports the presence of a primary stable remanent magnetism carried mostly by single-domain to multidomain magnetite/maghemite of late Miocene to Pliocene age, confirming that the widespread late Cenozoic shallow-water carbonates are amenable for paleomagnetic dating. The results demonstrate that a legible magnetostratigraphy can be established despite mineralogical transformation (i.e., dolo-

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mitization) but also after uplift and emersion. The idea of a synchronous rhodoliths/corals shift around the Miocene-Pliocene boundary at a Pacific-wide scale and possibly elsewhere could be verified by using magnetostratigraphy on similar transitions from other locations. It is significant that not only "quiet," subsiding carbonate platforms or atolls can be magnetostratigraphically dated and correlated, but the technique is also applicable to geodynamically active carbonate environments. This would be of particular interest for the active margin investigations if the technique can be replicated and extended to other uplifted carbonates.

Acknowledgments. The authors thank the New Caledonia ORSTOM scientists and technicians for their help in the field, and our friends from Maré Island, who offered us their hospitality. Thanks are extended to F.-J. Sierro for planktic foraminifera determinations, to the CFR laboratory for micromag facilities. J.E. Channel, E.L. Winterer, and a third, anonymous reviewer added constructive comments. The study was financially supported by the French Centre National de la Recherche Scientifique and by the U.S. National Science Foundation (grant INT 9116348 to D.F.M.). The RSMAS/University of Miami Paleomagnetic Laboratory is supported by the U.S. National Science Foundation and the W.M. Keck Foundation. CNRS-INSU Program DBT Contribution 741.

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(Received January 19, 1995; revised September 11, 1995; accepted September 15, 1995.)

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